

# PRANAV SINGH

Full Stack Developer | React.js | Next.js | Node.js | TypeScript

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## PROFESSIONAL SUMMARY

Full stack developer with hands-on experience across React.js, Next.js, Node.js, and Python backends. Delivered and deployed 3 production projects — a collaborative Kanban platform, a low-latency voice AI system, and a multi-provider AI chat app — each backed by measured performance benchmarks. Completed a 6-month frontend internship.

## TECHNICAL SKILLS

**Languages:** C++, JavaScript, TypeScript, Python, SQL

**Frontend:** HTML5, CSS3, React.js, Next.js, Tailwind CSS

**Backend:** Node.js, Flask, FastAPI, WebSockets

**Databases:** MySQL, Firebase, Firestore, Realtime Database

**AI & LLM:** OpenAI API, Anthropic API, Gemini API, Prompt Engineering

**Tools:** Git, GitHub, GitHub Actions, CI/CD, Vercel

## EXPERIENCE

### Frontend Developer Intern — Hackiware

Sep 2025 – Mar 2026

Live Site: [hackiware.com](https://hackiware.com) | In Development: [hacki-green.vercel.app](https://hacki-green.vercel.app)

- Owned frontend development end-to-end as the sole developer, building the site from scratch over a 6-month engagement.
- Maintained 2 parallel builds — live production and staging — shipping features without disrupting the public site.

## PROJECTS

### Sprinto — Real-Time Collaborative Workspace & Kanban Platform

Jul 2026 – Aug 2026

Tech: Next.js, TypeScript, Firebase, Tailwind CSS, @dnd-kit, IndexedDB | Live Demo: [sprinto-move.vercel.app](https://sprinto-move.vercel.app) | GitHub: [github.com/PRANAV-SINGH-CSE/Sprinto](https://github.com/PRANAV-SINGH-CSE/Sprinto)

- Developed a real-time Kanban platform supporting simultaneous multi-user editing, with <1ms local updates and ~4.5ms median cross-client sync.
- Cut repeat-visit board load time from ~400ms to ~5ms using an IndexedDB local cache layer.
- Averaged 345–400ms per Firestore write across CRUD operations, with full board provisioning (4 default columns) completing in ~1.8s.
- Automated presence tracking via 60-second heartbeat cycles and delivered chat messages at ~1.6ms median (9ms p95).
- Tuned drag-and-drop gesture recognition to ~12.5ms using a 5px pointer activation threshold.

### LiveVoice AI — Real-Time Speech Intelligence & Desktop HUD Platform

Jun 2026 – Jul 2026

Tech: Python, FastAPI, WebSockets, AsyncIO, PyQt5, OpenAI API, Anthropic API, Faster-Whisper

- Engineered a voice assistant achieving sub-700ms latency from end of speech to first spoken token, with a sub-1-second full response loop.
- Ran speech-to-text at a 0.28x real-time factor (3.5x faster than speech) while streaming LLM output at ~197 tokens/sec.
- Implemented a sub-12μs NLP filtering layer handling up to 93,000 queries/sec for real-time question and hallucination detection.
- Processed audio DSP (RMS, resampling, voice-activity detection) in under 30μs, sustaining 200,000+ chunks/sec.
- Serialized WebSocket messages at ~2μs per operation, sustaining 400,000+ messages/sec between backend and desktop overlay.

## ZentiqAI — AI Chat Web Application

Feb 2026 – Apr 2026

Tech: Python (Flask), JavaScript, Firebase | Live Demo: [zentiqai.vercel.app](https://zentiqai.vercel.app) | GitHub: [github.com/PRANAV-SINGH-CSE/ZentiqAI\\_Chat\\_Bot](https://github.com/PRANAV-SINGH-CSE/ZentiqAI_Chat_Bot)

- Architected a hybrid platform routing across 3 inference providers (Groq, Gemini, Ollama), achieving <600ms cloud response times.
- Streamlined an async FIFO dispatch queue at ~7ms median latency and a multimodal pipeline decoding 1080p images in <0.05ms.
- Deployed a Vercel–Firebase frontend with a FastAPI backend achieving 1.37ms p50 API latency and 1.27–1.64ms session response times.
- Integrated offline inference (Qwen 2.5/LLaVA) at 1.5–3.2s generation time, reaching 300+ tokens/sec cloud streaming.

## HACKATHONS

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### Adobe University Hackathon 2026 — Adobe

- Selected among 3,01,000+ registrants nationwide; cleared Round 1 as part of a 2–3 member team, awaiting Round 2 results.

### Student Freelancing Platform — Smart India Hackathon (SIH)

Tech: Flutter, Firebase

- Collaborated in a team to develop a Flutter-based freelancing platform with 4 core workflows: profiles, listings, freelancer discovery, and job applications.
- Progressed through Round 1 of SIH while contributing to prototyping, feature integration, and debugging.

## CERTIFICATIONS

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- Full Stack Development Using Next.js — CipherSchools (Completed Aug 2026)
- Database Management System — Part 1 — Infosys Springboard (Issued Jul 2026)
- Data Structures & Algorithms — Lovely Professional University / iamneo (Issued Jun 2026)
- Object-Oriented Programming — Lovely Professional University / iamneo (Issued Jun 2026)
- Computer Programming (72 hours) — Lovely Professional University / iamneo (Issued May 2025)

## EDUCATION

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### Lovely Professional University (LPU)

B.Tech in Computer Science and Engineering | TGPA: 8.4

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Operating Systems

### Divine Grace Public School

Class 12 | Percentage: 66%

Class 10 | Percentage: 82%